

Analysis of Variance (ANOVA)

Hypothesis Testing in ANOVA (Partitioning of Variance in the ANOVA)

Null Hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3 \dots \mu_k$

Alternate Hypothesis H_1 : Alternate one of the sample mean is not equal.

we can't use this

$$\mu_1 \neq \mu_2 \neq \mu_3 \dots \mu_k$$

Test statistics

$$F = \frac{\text{Variance between sample}}{\text{Variance within sample}}$$

σ^2

	α_1	α_2	α_3
variance within samples	1	6	5
	2	1	6
	4	3	3
	5	2	2
	3	1	4
	$\overline{\alpha_1} = 3$	$\overline{\alpha_2} = 19/5$	$\overline{\alpha_3} = 4$

$$H_0 = \bar{x}_1 = \bar{x}_2 = \bar{x}_3$$

H_1 At least one sample mean is not equal

One way ANOVA

One factor with atleast 2 levels, levels are independent.

- ① Doctor want to test a new medication which reduces headache. They split the participant into 3 condition [15mg, 30mg, 45mg]. later on the doctor ask the patient to rate the headache between [1-10]. Are there any differences between the 3 conditions $\alpha = 0.05$?

Ans →

15 mg	30mg	45mg
9	7	4
8	6	3
7	6	2
8	7	3
8	8	4
9	7	3
8	6	2

① Define Null and Alternate Hypothesis ?

$$H_0 : \mu_{15} = \mu_{30} = \mu_{45}$$

H_1 : not all μ are equal

② Significance

$$\alpha = 0.05$$

$$C.D = 0.95$$

③ Calculate Degree of freedom

$N = 21$ 3 columns $\rightarrow$ multiply them
7 row
 $a = 3$ $n = 7$

$$df_{\text{between}} = a - 1 = 3 - 1 = 2$$

$$df_{\text{within}} = N - a = 21 - 3 = 18$$

$$df_{\text{total}} = N - 1 = 20$$

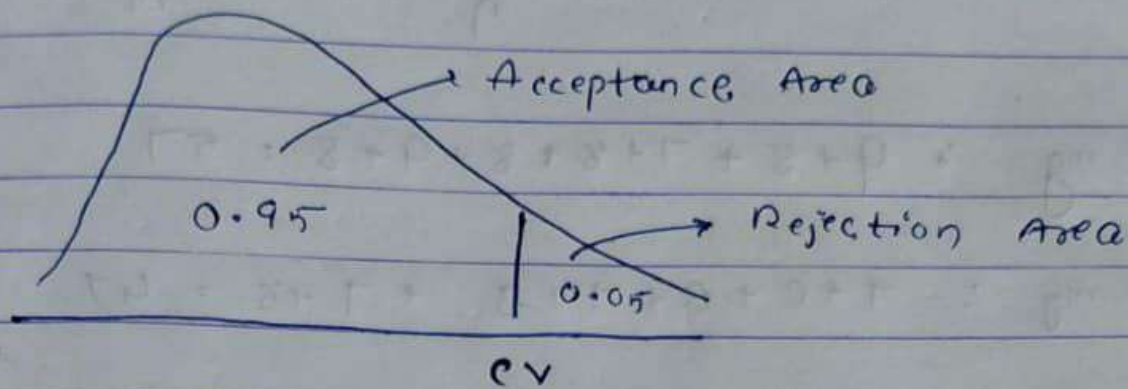
α, α_c
 $(2, 18)$
 $\downarrow$

F-table

$$\alpha = 0.05$$

$\downarrow$
critical value

(a) Decision Boundary



3.5546

→ using F Table

Decision Rule

If F is greater than 3.5546, reject the Null Hypothesis

(b) Calculate F Test statistics

	SS	df	MS	F
Between	98.67			
within	10.29			
Total	108.96			

∴ SS → sum of Square

MS → Mean square

$$\textcircled{1} \text{ SS between} = \frac{\sum (Z a_i)^2}{n} - \frac{T^2}{n}$$

$$15 \text{ mg} : 9 + 8 + 7 + 8 + 8 + 9 + 8 = 57$$

$$30 \text{ mg} : 7 + 6 + 6 + 7 + 8 + 7 + 6 = 47$$

$$45 \text{ mg} : 4 + 3 + 2 + 3 + 4 + 3 + 2 = 21$$

$$= \frac{57^2 + 47^2 + 21^2}{7} - \frac{[57^2 + 47^2 + 21^2]}{21}$$

$$= \boxed{98.67}$$

$$\textcircled{2} \text{ SS within} = \sum y^2 - \frac{\sum (a_i)^2}{n}$$

$$\sum y^2 = 9^2 + 8^2 + 7^2 + 8^2 + 8^2 + 9^2 + 8^2$$

$$= 853$$

$$= 853 - \frac{[57^2 + 47^2 + 21^2]}{7}$$

$$= \boxed{10.29}$$

55

	SS	df	MS	F
Between	98.67	2	49.34	
Within	10.29	18	0.57	
Total	108.96	20		

$$F \text{ test} = \frac{MS \text{ Between}}{MS \text{ within}}$$

$$F = \frac{\text{Variance between sample}}{\text{Variance within sample}}$$

$$F = \frac{49.34}{0.57} = \underline{\underline{86.56}}$$

If F is greater than 3.5546, Reject the H_0

$$86.56 > 3.5546 \quad \text{Reject the } H_0$$